

Do grants improve firms' energy efficiency?

Evidence from a randomized controlled trial with 1,964 European SMEs under the EENergy pilot

Summary of the final report, 2026. Estimates are drawn from the report's main findings and supporting analyses.

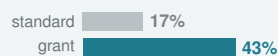
Design	Sample	Coverage	Policy	Registry
Randomized controlled trial	1,964 SMEs: 986 offered the grant, 978 standard group	34 European countries	Grant of up to EUR 10,000 on top of standard advisory services	AEARCTR-0015124

SUMMARY

The grant offer changed **what firms bought** and **how they manage energy practices**. It roughly **doubled** the share of firms investing in energy efficiency, **tripled** the share investing in renewables, and raised energy monitoring and goal-setting. Electricity savings appear only among micro and small firms and in Call 2.

+26 pp

43% of firms in the grant group **made an energy-efficiency investment** against 17% of firms in the standard group



+20 pp

Invested in renewable energy: From 10% to about 30% — the share roughly tripled



+EUR 9,700

Additional total energy-related investment per firm assigned to the grant group — nearly double the EUR 10,100 invested by firms in the standard group

not detected

Reduction in energy consumption (on average); **improvement of overall business performance**

How the study worked

The EENergy pilot, funded by the EU Single Market program, offered SMEs a reimbursement grant of up to EUR 10,000 for energy-efficiency measures — equipment and installations, consultancy, or staff training — on top of the advisory support already provided by Enterprise Europe Network (EEN) advisors.

Firms applied in two calls (February–May 2024 and December 2024–February 2025) and were randomly assigned by lottery to a **grant group**, offered the grant alongside standard advisory services, or a **standard group**, which received advisory services only. The two groups were closely balanced (similar) at application stage on employment, revenue, energy expenditure, firm size, as well as consortium and advisor characteristics. Later differences between them can therefore be attributed to the grant offer.

Who took part

54% micro, **36%** small and **9%** medium-sized firms; median **8** full-time employees, median revenue about **EUR 650,000**, median annual energy costs about **EUR 12,700** (2.4% of annual revenue). **36%** were in manufacturing; **62%** already monitored their energy use, but only **22%** had invested in energy efficiency in the year before the program.



Participating firms are larger and far more manufacturing-intensive than the average EU SME — see “How to read the following numbers” on this page.

Timeline



Further follow-up data collection is planned.

Take-up and the weight of the grant

- Winning the lottery raised the probability of **receiving a grant by 81 percentage points**. Considering that not all firms assigned the grant requested reimbursements, the average grant received was about **EUR 7,700**.
- Total energy-related investment in the grant group was about **EUR 9,700 higher** per firm than in the standard group, suggesting that firms contributed some additional spending beyond the reimbursement rather than simply replacing private expenditure one-for-one.
- About one in five** firms assigned the grant never signed the contract and non-signers were disproportionately larger firms.
- Firms of all sizes claimed close to the cap, but its relative weight differed sharply: The fixed EUR 10,000 was worth about **5.1% of annual revenue** for the median micro firm (up to about 40% for the smallest), against about **0.1%** for the median medium-sized firm.
- The grant **partly substituted for other public support**: Other public co-financing was about 1.4% in the grant group, compared with 5.4% in the standard group.

How to read the following numbers

- Differential survey response.** Roughly 22% of enrolled firms provided no follow-up information, and the survey response differed sharply by group (91% of the grant group against 65% of the standard group). All main estimates correct for these differences in response rate and are robust to various assumptions about the non-responders.
- Intent-to-treat.** Effects are those of *being offered* the grant, not of using it (effects among firms that actually used the grant are larger).

What firms did with the grant

- The grant offer roughly **doubled** the share of firms making an energy-efficiency investment (17% to about 43%) and **tripled** the share investing in renewable energy (10% to about 30%). Measured cumulatively across all available follow-up rounds, the response is even larger: Energy-efficiency investment rose from 25% in the standard group to about 58% in the grant group (+33 pp), and renewable-energy investment from 15% to about 42% (+27 pp).
- The additional spending went **overwhelmingly toward physical investment**: Capital expenditure roughly doubled, rising by about **EUR 9,300**, compared with much smaller changes in consultancy and training expenditure. Renewable-energy installations account for about **half** of the additional capital spending (about EUR 4,924), followed by smaller increases in machinery, HVAC, lighting and building improvements.

Firm size mattered for whether firms invested

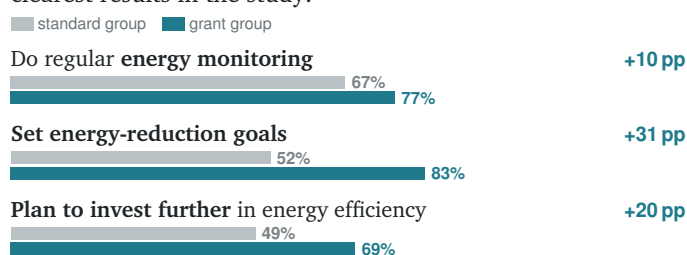
Renewable-energy investment rose by **22 pp** among micro and small firms, against **5 pp** among medium-sized firms — a statistically significant difference ($p = 0.015$).

Energy mix and use

- The **composition** of energy use shifted towards renewables: Firms in the grant group were **7.7 pp** more likely to report using renewable energy, against a standard-group rate of 15% — an increase of roughly 50%.
- For the full sample, there is **no detectable reduction in electricity consumption or total energy expenditure** one year after the grants. Using later available observations, however, electricity spending falls by about **10%** ($p < 0.05$).
- Evidence of electricity savings** was found for specific groups of firms: Electricity consumption fell by about **10%** among micro and small firms; electricity spending fell by about **20%** among firms from Call 2.

How firms manage energy practices

Beyond buying equipment, firms in the grant group changed how they managed energy practices — these are among the clearest results in the study:



These changes were broadly similar across firm sizes. Engagement with EEN advisors also increased substantially among firms in the grant group compared with the standard group, and by similar amounts across sizes. Firms in the grant group also expressed modestly higher agreement with climate-action statements.

What might explain the pattern

The report examines different candidate explanations for differences in impacts across groups. The heterogeneity patterns point most clearly to one of them:

Relative size of the support

The fixed EUR 10,000 was worth about **5% of annual revenue** for the median micro firm, against **0.1%** for the median medium-sized firm. Consistent with this, firms with below-median baseline revenue increased renewable-energy investment by **26 pp**, against **14 pp** for higher-revenue firms ($p = 0.016$). **Caveat:** This interpretation remains **suggestive**; the comparisons do not experimentally isolate the mechanism behind the stronger response among smaller firms.

Business performance

The program produced **no detectable improvement in overall business performance at the one-year follow-up**. There are no statistically significant effects on revenue, profits, employment, or the value of machinery and equipment. The estimated revenue effect is positive but imprecise and cannot be distinguished from zero, including after adjustment for multiple testing. This conclusion holds across the revenue distribution and under the alternative outcome definitions and robustness checks.

What the study did *not* find on average

- ✗ No detectable reduction in electricity consumption for the grant group.
- ✗ No detectable reduction in total energy expenditure at one year.
- ✗ No detectable improvement in revenue, profits, employment, or the value of machinery and equipment.
- ✗ No differences in impacts by the levels of baseline liquidity, managerial practices, or energy intensity at the baseline.

“No detectable effect” means the estimate cannot be statistically distinguished from zero at the one-year horizon — not that the true effect is necessarily zero.

Policy recommendations

- Target micro and small firms.** A fixed EUR 10,000 was ~5% of revenue for a median micro firm but ~0.1% for a medium firm; renewable investment rose 22 pp vs. 5 pp. Scale the cap to firm size or restrict eligibility.
- Justify the grant on investment and management change, not business performance.** It doubled energy-efficiency investment, tripled renewable investment and raised monitoring and goal-setting, but had no detectable effect on revenue, profits or employment at one year after the grant awards.